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Packet Tracer - Configure Layer 3 Switching and Inter-VLAN Routing

Addressing Table

Device	Interface	IP Address / Prefix
MLS	VLAN 10	192.168.10.254 /24
		2001:db8:acad:10::1/64
	VLAN 20	192.168.20.254 /24
		2001:db8:acad:20::1/64
	VLAN 30	192.168.30.254/24
		2001:db8:acad:30::1/64
	VLAN 99	192.168.99.254/24
	G0/2	209.165.200.225
		2001:db8:acad:a::1/64
PC0	NIC	192.168.10.1
PC1	NIC	192.168.20.1
PC2	NIC	192.168.30.1
PC3	NIC	192.168.10.2/24
		2001:db8:acad:10::2/64
PC4	NIC	192.168.20.2/24
		2001:db8:acad:20::2/64
PC5	NIC	192.168.30.2
		2001:db8:acad:30::2/64
S1	VLAN 99	192.168.99.1
S2	VLAN 99	192.168.99.2

Device	Interface	IP Address / Prefix
S3	VLAN 99	192.168.99.3

Objectives

Part 1: Configure Layer 3 Switching

Part 2: Configure Inter-VLAN Routing

Part 3: Configure IPv6 Inter-VLAN Routing

Background / Scenario

A multilayer switch like the Cisco Catalyst 3650 is capable of both Layer 2 switching and Layer 3 routing. One of the advantages of using a multilayer switch is this dual functionality. A benefit for a small to medium-sized company would be the ability to purchase a single multilayer switch instead of separate switching and routing network devices. Capabilities of a multilayer switch include the ability to route from one VLAN to another using multiple switched virtual interfaces (SVIs), as well as the ability to convert a Layer 2 switchport to a Layer 3 interface.

Instructions

Part 1: Configure Layer 3 Switching

In Part 1, you will configure the GigabitEthernet 0/2 port on switch MLS as a routed port and verify that you can ping another Layer 3 address.

- On MLS, configure G0/2 as a routed port and assign an IP address according to the Addressing Table.

```
MLS(config)# interface g0/2
MLS(config-if)# no switchport
MLS(config-if)# ip address 209.165.200.225 255.255.255.252
```

- Verify connectivity to **Cloud** by pinging 209.165.200.226.

```
MLS# ping 209.165.200.226
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

```
MLS#ping 209.165.200.226

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

MLS#
```

Part 2: Configure Inter-VLAN Routing

Step 1: Add VLANs.

Add VLANs to MLS according to the table below. Packet Tracer scoring is case-sensitive, so type the names exactly as shown.

VLAN Number	VLAN Name
10	Staff
20	Student
30	Faculty

```
MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLS(config)#vlan 10
MLS(config-vlan)#name Staff
MLS(config-vlan)#vlan 20
MLS(config-vlan)#name Student
MLS(config-vlan)#vlan 30
MLS(config-vlan)#name Faculty
MLS(config-vlan)#exit
MLS(config)#
MLS(config)#
```

Step 2: Configure SVI on MLS.

Configure and activate the SVI interfaces for VLANs 10, 20, 30, and 99 according to the Addressing Table. The configuration for VLAN 10 is shown below as an example.

```
MLS(config)# interface vlan 10
MLS(config-if)# ip address 192.168.10.254 255.255.255.0
```

```
MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLS(config)#
MLS(config)#interface vlan 10
MLS(config-if)#ip address 192.168.10.254 255.255.255.0
MLS(config-if)#ipv6 address 2001:db8:acad:10::1/64
MLS(config-if)#exit
MLS(config)#
MLS(config)#interface vlan 20
MLS(config-if)#ip address 192.168.20.254 255.255.255.0
MLS(config-if)#ipv6 address 2001:db8:acad:20::1/64
MLS(config-if)#exit
MLS(config)#interface vlan 30
MLS(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up
MLS(config-if)#ip address 192.168.30.254 255.255.255.0
MLS(config-if)#ipv6 address 2001:db8:acad:30::1/64
MLS(config-if)#interface vlan 99
MLS(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up
MLS(config-if)#ip address 192.168.99.254 255.255.255.0
MLS(config-if)#exit
MLS(config)#
```

Step 3: Configure Trunking on MLS.

Trunk configuration differs slightly on a Layer 3 switch. On the Layer 3 switch, the trunking interface needs to be encapsulated with the dot1q protocol, however it is not necessary to specify VLAN numbers as it is when working with a router and subinterfaces.

- On MLS, configure interface **g0/1**.
- Make the interface a static trunk port.

```
MLS(config-if)# switchport mode trunk
```

- Specify the native VLAN as 99.

```
MLS(config-if)# switchport trunk native vlan 99
```

- Encapsulate the link with the dot1q protocol.

```
MLS(config-if)# switchport trunk encapsulation dot1q
```

Note: Packet Tracer may not score the trunk encapsulation.

```
MLS(config)#interface g0/1
MLS(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured
to "trunk" mode.
MLS(config-if)#switch
MLS(config-if)#switchport trunk native vlan 99
MLS(config-if)#switch
MLS(config-if)#switchport trunk encapsulation dot1q
MLS(config-if)#
```

Step 4: Configure trunking on S1.

- Configure interface **g0/1** of S1 as a static trunk.
- Configure the native VLAN on the trunk.

```
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int g0/1
S1(config-if)#switch
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

S1(config-if)#switch
S1(config-if)#switchport trunk native vlan 99
S1(config-if)#exit
S1(config)#
```

Step 5: Enable routing.

- Use the **show ip route** command.

Are there any active routes? **No**

```
MLS#show ip route
Default gateway is not set

Host          Gateway          Last Use    Total Uses  Interface
ICMP redirect cache is empty

MLS#
```

- b. Enter the **ip routing** command to enable routing in global configuration mode.

```
MLS(config)# ip routing
```

- c. Use the **show ip route** command to verify routing is enabled.

```
MLS# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.99.0/24 is directly connected, Vlan99
     209.165.200.0/30 is subnetted, 1 subnets
C      209.165.200.224 is directly connected, GigabitEthernet0/2
```

```
MLS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

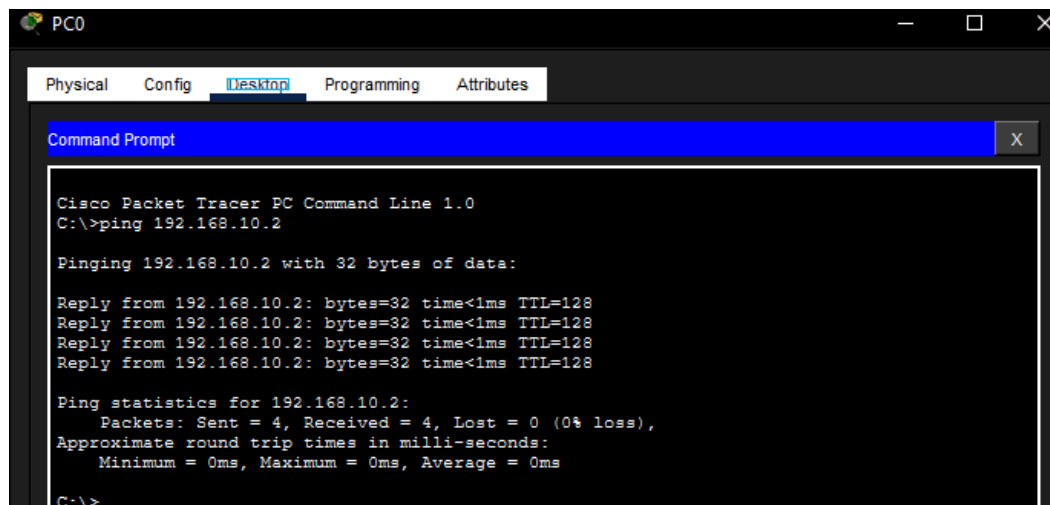
Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.99.0/24 is directly connected, Vlan99
     209.165.200.0/30 is subnetted, 1 subnets
C      209.165.200.224 is directly connected, GigabitEthernet0/2

MLS#
```

Step 6: Verify end-to-end connectivity.

- a. From PC0, ping PC3 or MLS to verify connectivity within VLAN 10.



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

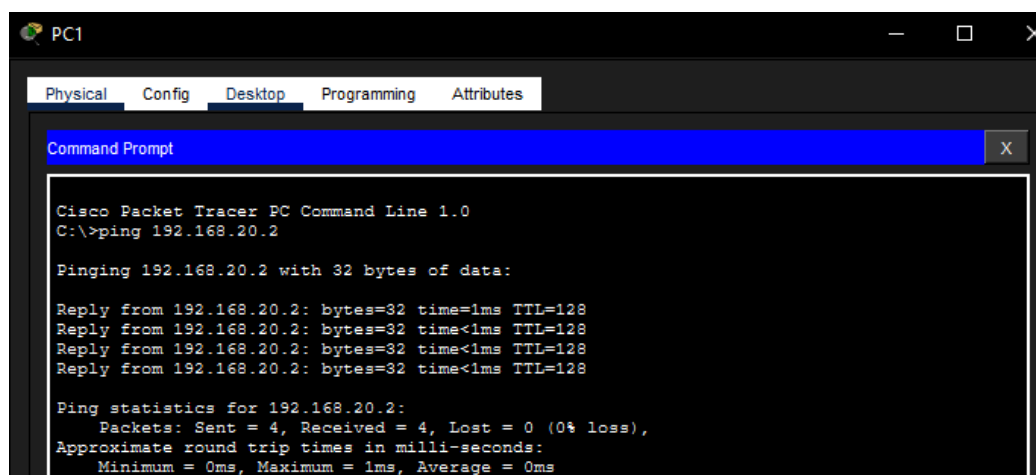
Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

b. From PC1, ping PC4 or MLS to verify connectivity within VLAN 20.



PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

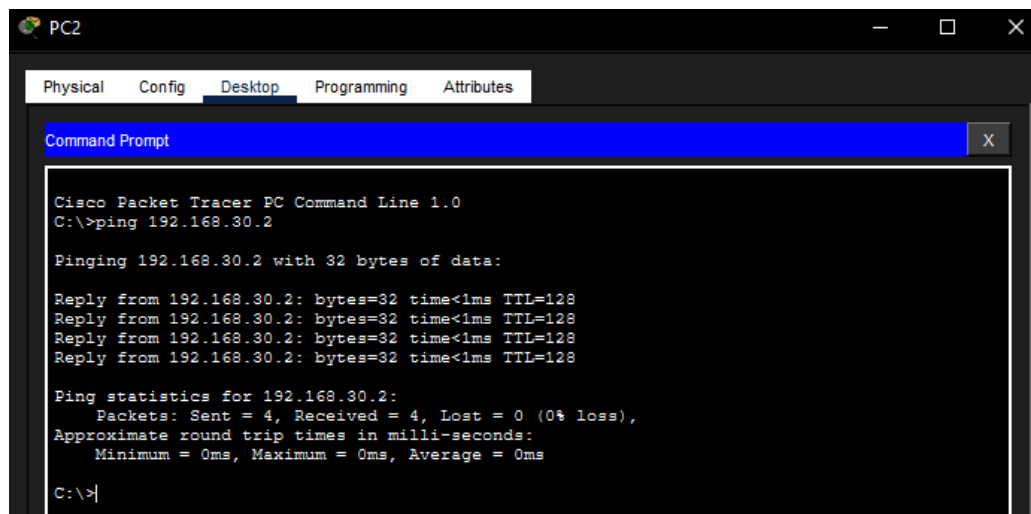
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

c. From PC2, ping PC5 or MLS to verify connectivity within VLAN 30.



PC2

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

d. From S1, ping S2, S3, or MLS to verify connectivity with VLAN 99.

```
S1#ping 192.168.99.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

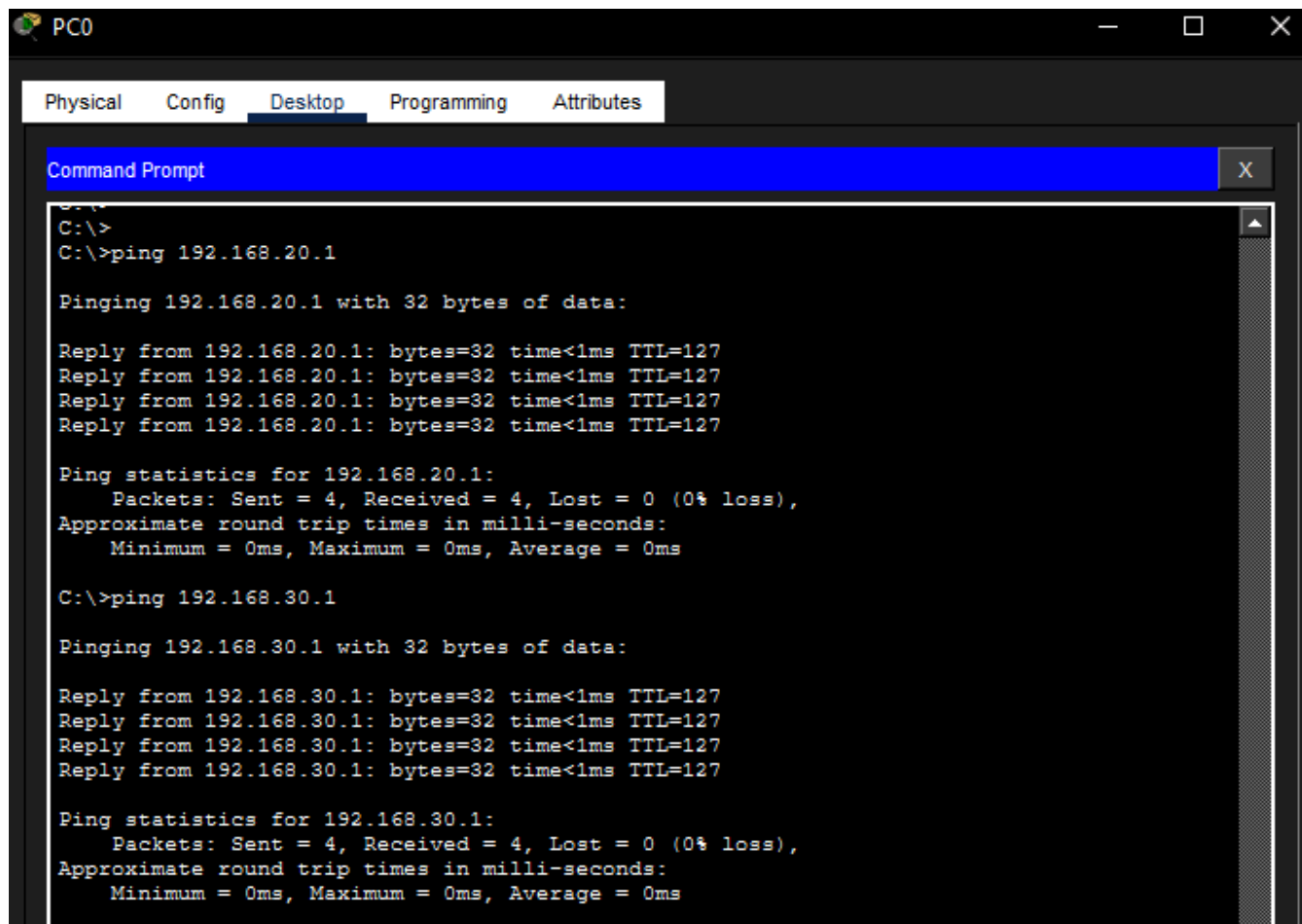
S1#ping 192.168.99.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

S1#ping 192.168.99.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.254, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

- e. To verify inter-VLAN routing, ping devices outside the sender's VLAN.



- f. From any device, ping this address inside **Cloud**, 209.165.200.226.

The Layer 3 switch is now routing between VLANs and providing routed connectivity to the cloud.

PC1

```
C:\>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time<1ms TTL=254
Reply from 209.165.200.226: bytes=32 time<1ms TTL=254
Reply from 209.165.200.226: bytes=32 time<1ms TTL=254
Reply from 209.165.200.226: bytes=32 time<1ms TTL=254

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Part 3: Configure IPv6 Inter-VLAN Routing

Layer 3 switches also route between IPv6 networks.

Step 1: Enable IPv6 routing.

Enter the **ipv6 unicast-routing** command to enable IPv6 routing in global configuration mode.

```
MLS(config)# ipv6 unicast-routing
```

Step 2: Configure SVI for IPv6 on MLS.

Configure IPv6 addressing on SVI for VLANs 10, 20, and 30 according to the Addressing Table. The configuration for VLAN 10 is shown below.

```
MLS(config)# interface vlan 10
MLS(config-if)# ipv6 address 2001:db8:acad:10::1/64
```

Step 3: Configure G0/2 with IPv6 on MLS.

a. Configure IPv6 addressing on G0/2.

```
MLS(config)# interface G0/2
MLS(config-if)# ipv6 address 2001:db8:acad:a::1/64
```

```
MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLS(config)#ipv6 unicast-routing
MLS(config)#
MLS(config)#interface vlan 10
MLS(config-if)#ipv6 address 2001:db8:acad:10::1/64
MLS(config-if)#exit
MLS(config)#interface vlan 20
MLS(config-if)#ipv6 address 2001:db8:acad:20::1/64
MLS(config-if)#exit
MLS(config)#interface vlan 30
MLS(config-if)#ipv6 address 2001:db8:acad:30::1/64
MLS(config-if)#exit
MLS(config)#
MLS(config)#interface g0/2
MLS(config-if)#ipv6 address 2001:db8:acad:a::1/64
MLS(config-if)#exit
MLS(config)#
```


- b. Use the **show ipv6 route** command to verify IPv6 connected networks.

```
MLS# show ipv6 route
```

```
IPv6 Routing Table - 10 entries
```

```
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
```

```
U - Per-user Static route, M - MIPv6
```

```
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
```

```
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
```

```
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
```

```
D - EIGRP, EX - EIGRP external
```

```
S  ::/0 [1/0]
    via 2001:DB8:ACAD:A::2, GigabitEthernet0/2
C  2001:DB8:ACAD:A::/64 [0/0]
    via ::, GigabitEthernet0/2
L  2001:DB8:ACAD:A::1/128 [0/0]
    via ::, GigabitEthernet0/2
C  2001:DB8:ACAD:10::/64 [0/0]
    via ::, Vlan10
L  2001:DB8:ACAD:10::1/128 [0/0]
    via ::, Vlan10
C  2001:DB8:ACAD:20::/64 [0/0]
    via ::, Vlan20
L  2001:DB8:ACAD:20::1/128 [0/0]
    via ::, Vlan20
C  2001:DB8:ACAD:30::/64 [0/0]
    via ::, Vlan30
L  2001:DB8:ACAD:30::1/128 [0/0]
    via ::, Vlan30
L  FF00::/8 [0/0]
    via ::, Null0
```

```

MLS#show ipv6 route
IPv6 Routing Table - 10 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
S   ::/0 [1/0]
    via 2001:DB8:ACAD:A::2
C   2001:DB8:ACAD:A::/64 [0/0]
    via ::, GigabitEthernet0/2
L   2001:DB8:ACAD:A::1/128 [0/0]
    via ::, GigabitEthernet0/2
C   2001:DB8:ACAD:10::/64 [0/0]
    via ::, Vlan10
L   2001:DB8:ACAD:10::1/128 [0/0]
    via ::, Vlan10
C   2001:DB8:ACAD:20::/64 [0/0]
    via ::, Vlan20
L   2001:DB8:ACAD:20::1/128 [0/0]
    via ::, Vlan20
C   2001:DB8:ACAD:30::/64 [0/0]
    via ::, Vlan30
L   2001:DB8:ACAD:30::1/128 [0/0]
    via ::, Vlan30
L   FF00::/8 [0/0]
    via ::, Null0
MLS#
MLS#

```

Step 4: Verify IPv6 connectivity.

Devices PC3, PC4, and PC5 have been configured with IPv6 addresses. Verify IPv6 inter-VLAN routing and connectivity to **Cloud**.

- a. From PC3, ping MLS to verify connectivity within VLAN 10.

```

PC3
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:acad:10::1

Pinging 2001:db8:acad:10::1 with 32 bytes of data:

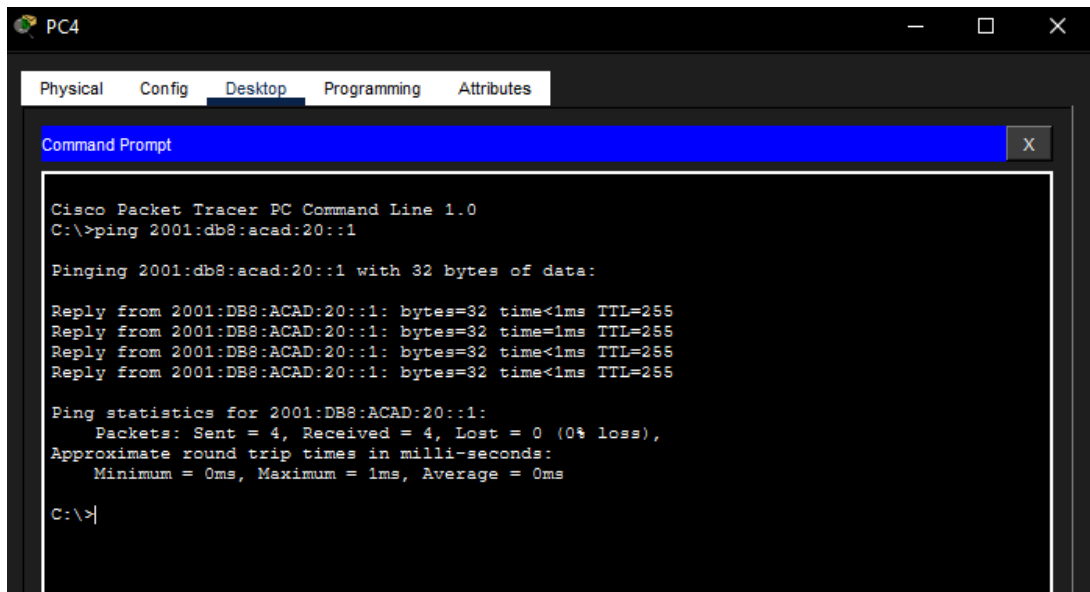
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:ACAD:10::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

- b. From PC4, ping MLS to verify connectivity within VLAN 20.



The screenshot shows the PC4 interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:acad:20::1

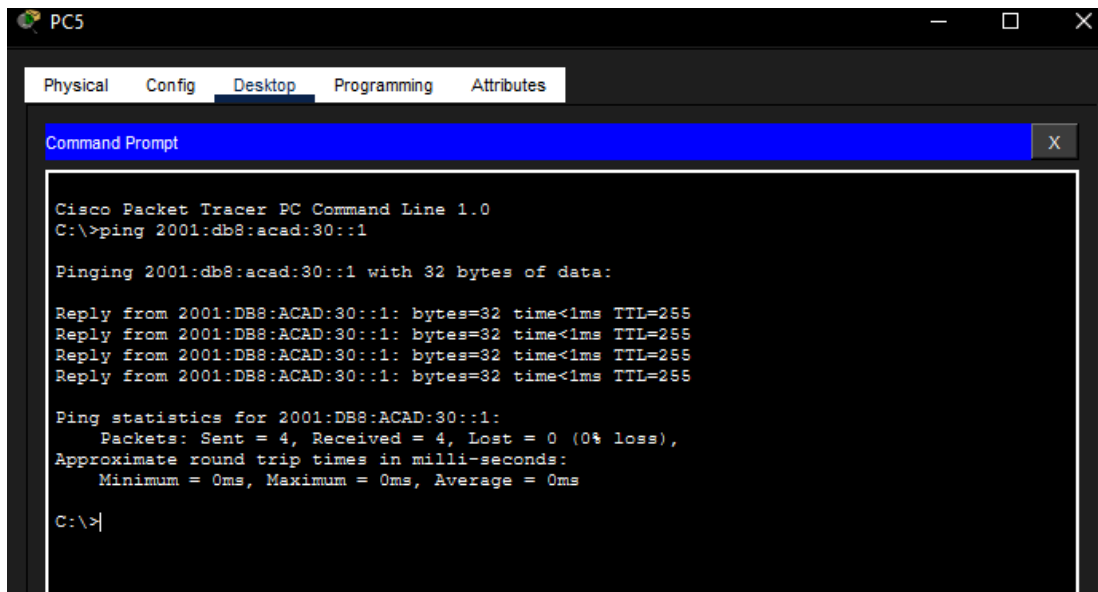
Pinging 2001:db8:acad:20::1 with 32 bytes of data:

Reply from 2001:DB8:ACAD:20::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:20::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:20::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:20::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:ACAD:20::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

- c. From PC5, ping MLS to verify connectivity within VLAN 30.



The screenshot shows the PC5 interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:acad:30::1

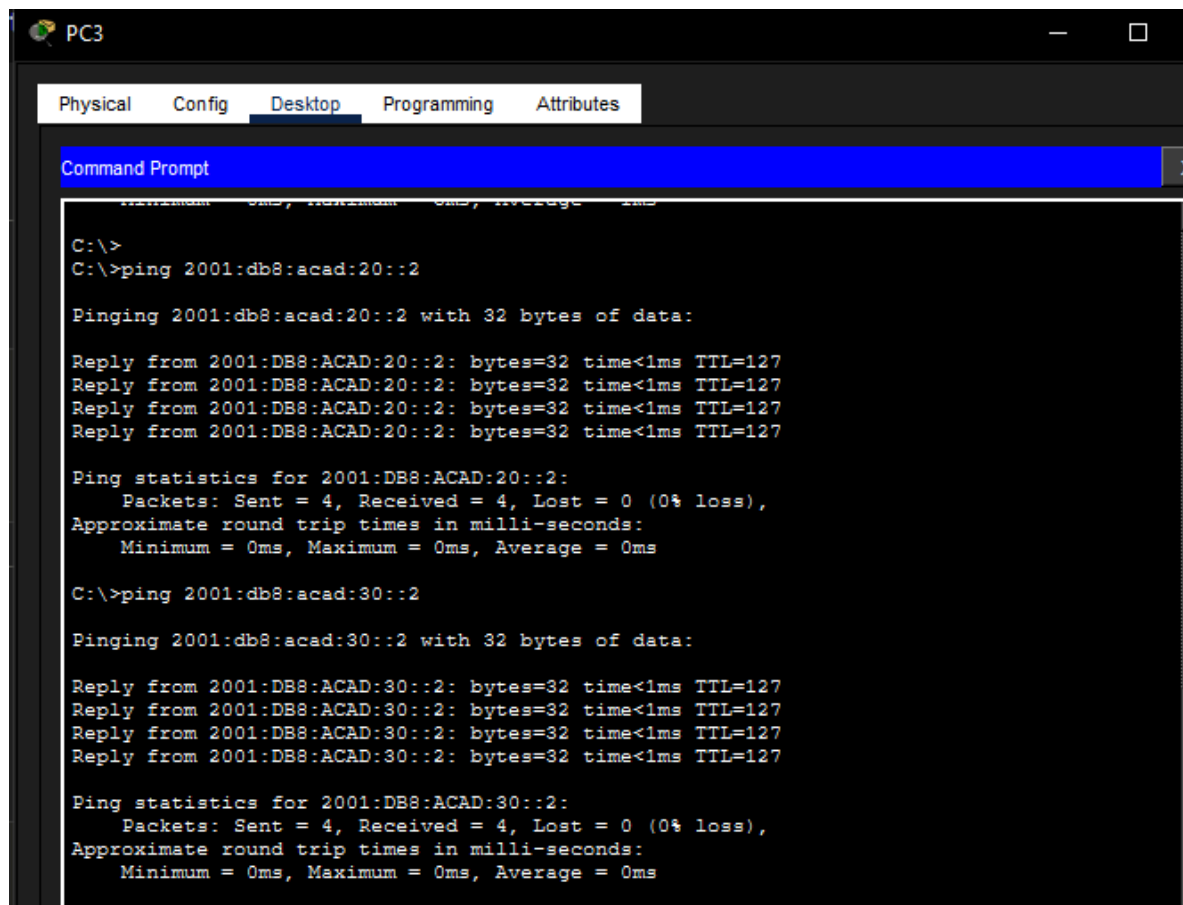
Pinging 2001:db8:acad:30::1 with 32 bytes of data:

Reply from 2001:DB8:ACAD:30::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:30::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:30::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:30::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:ACAD:30::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

- d. To verify inter-VLAN routing, ping between devices PC3, PC4, and PC5.



The screenshot shows the PC3 interface with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the following text:

```
C:\>
C:\>ping 2001:db8:acad:20::2

Pinging 2001:db8:acad:20::2 with 32 bytes of data:

Reply from 2001:DB8:ACAD:20::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::2: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:20::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

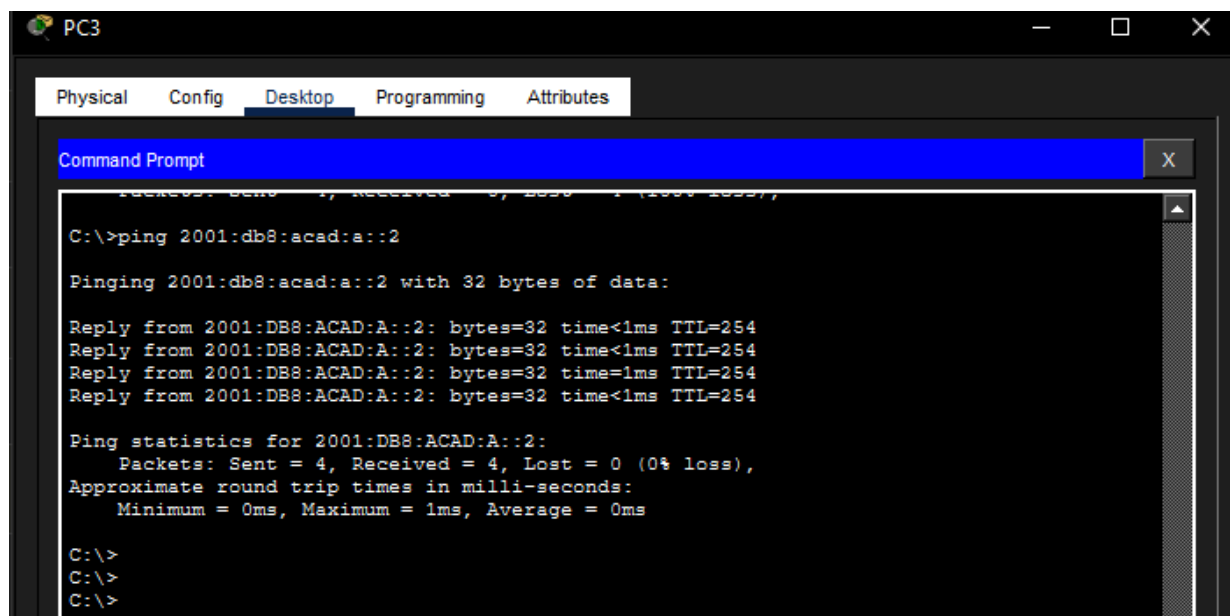
C:\>ping 2001:db8:acad:30::2

Pinging 2001:db8:acad:30::2 with 32 bytes of data:

Reply from 2001:DB8:ACAD:30::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::2: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:30::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- e. From PC3 ping the address inside **Cloud**, 2001:db8:acad:a::2.



The screenshot shows the PC3 interface with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the following text:

```
C:\>ping 2001:db8:acad:a::2

Pinging 2001:db8:acad:a::2 with 32 bytes of data:

Reply from 2001:DB8:ACAD:A::2: bytes=32 time<1ms TTL=254
Reply from 2001:DB8:ACAD:A::2: bytes=32 time<1ms TTL=254
Reply from 2001:DB8:ACAD:A::2: bytes=32 time<1ms TTL=254
Reply from 2001:DB8:ACAD:A::2: bytes=32 time<1ms TTL=254

Ping statistics for 2001:DB8:ACAD:A::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>
C:\>
```